information needed for testing: the generated text and whether the response is complete. Ideally, you want to balance detail and simplicity in the mock responses to ensure effective testing without compromising flexibility.

8.4.3 Testing Vercel AI SDK responses

Let's now focus on how to test Vercel AI SDK functuality. The SDK provides tools that facilitate effective testing of applications using generative AI. It includes mock providers and test helpers that allow developers to simulate model behavior, at least with a predefined list of responses.

The key helpers that the framework exposes for testing are `MockLanguageModelV1`, a mock language model that allows you to define custom responses for testing, and `simulateReadableStream`, a helper function that simulates streaming responses so you can test streaming functionalities. Both helpers are useful for creating reliable and efficient tests that ensure the application behaves as expected under different conditions without using actual calls to an API.

Running the unit tests

To run these unit tests, navigate to the ch08/chat-testing directory in your terminal (where where the package.json file, which defines the test script, is located) and execute the `npm run test` command:

```
cd ch08/chat-testing
$ npm run test
```

MOCKLANGUAGEMODELV1

The `MockLanguageModelV1` is a mock provider included in the Vercel AI SDK that allows developers to simulate the behavior of a language model during testing. By using this mock, developers can define specific responses to test various scenarios without calling the provider API. When you need to mock one of the AI generator functions (`generateText`, `generateObject`) without calling the actual AI provider, you just need to create a mock instance of a model using the `MockLanguageModelV1` class with the dummy content as a response. The following listing shows the simplest use case.

Listing 8.7 Testing with `MockLanguageModelV1`

```
import { generateText } from 'ai';
import { MockLanguageModelV1 } from 'ai/test';

describe('Text Generation Tests', () => {
  test('should return predefined text from mock model', async () => {
    const result = await generateText({
      model: new MockLanguageModelV1({
```

Imports the MockLanguageModelV1 to simulate a language model for testing purposes

Creates a new instance of MockLanguageModelV1 to simulate a language model's behavior